

Euclidean Space

Coordinate Vectors - Subspaces

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Inner Products

1 Inner Products

2 Orthogonal Complements

3 Orthogonal Projection



Recall

Inner products

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos(\vec{a}, \vec{b})$$

- $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$
- $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$
- $(k\vec{a}) \cdot \vec{b} = (k\vec{b}) \cdot \vec{a} = k\vec{a} \cdot \vec{b}$



Standard Inner Product

Definition

Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$. Then the **standard inner product** of $\mathbf{u}$ and $\mathbf{v}$ is

$$(\mathbf{u}, \mathbf{v}) := \mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = \begin{pmatrix} u_1 & \dots & u_n \end{pmatrix} \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} = u_1 v_1 + \dots + u_n v_n.$$

Example Compute $\mathbf{u} \cdot \mathbf{v}$ for $\mathbf{u} = \begin{pmatrix} 2 \\ -5 \\ -1 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 3 \\ 2 \\ -3 \end{pmatrix}$.

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = \begin{pmatrix} 2 & -5 & -1 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \\ -3 \end{pmatrix} = 2 \times 3 - 5 \times 2 + (-1) \times (-3) = -1.$$



Properties

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be vectors in $\mathbb{R}^n$, c be a scalar, and $(\mathbf{u}, \mathbf{v})$ be the standard inner product in $\mathbb{R}^n$. Then

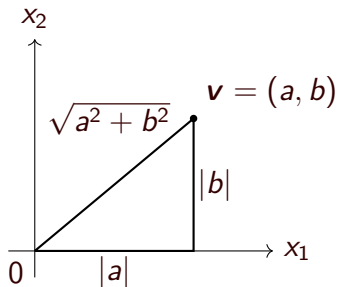
- a** Symmetry: $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
- b** Linearity: $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$
- c** Homogeneity: $(c\mathbf{u}) \cdot \mathbf{v} = c(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u} \cdot (c\mathbf{v})$
- d** Positive definiteness: $\mathbf{u} \cdot \mathbf{u} \geq 0$, and $\mathbf{u} \cdot \mathbf{u} = 0$ if and only if $\mathbf{u} = \mathbf{0}$



Definition

The **length** (or **norm**) of $\mathbf{v}$ is the nonnegative scalar $\|\mathbf{v}\|$ defined by

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}, \quad \text{and } \|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v}$$



- A vector whose length is 1 is called a **unit vector**.
- If we divide a nonzero vector $\mathbf{v}$ by its length, we obtain a unit vector $\mathbf{u}$
- The process of creating unit vector $\mathbf{u}$ from $\mathbf{v}$ is called **normalizing** $\mathbf{v}$, and we say that $\mathbf{u}$ is in the same direction as $\mathbf{v}$.



Example

Let $\mathbf{v} = (1, -2, 2, 0)$. Find a unit vector $\mathbf{u}$ in the same direction as $\mathbf{v}$.



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Compute the length of $\mathbf{v}$:

$$\|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v} = 1^2 + (-2)^2 + 2^2 + 0^2 = 9 \Rightarrow \|\mathbf{v}\| = 3.$$

Multiply $\mathbf{v}$ by $\frac{1}{\|\mathbf{v}\|}$ to obtain $\mathbf{u}$:

$$\mathbf{u} = \frac{1}{\|\mathbf{v}\|} \mathbf{v} = \frac{1}{3} \mathbf{v} = \frac{1}{3} \begin{pmatrix} 1 \\ -2 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1/3 \\ -2/3 \\ 2/3 \\ 0 \end{pmatrix}$$

Check that $\|\mathbf{u}\| = 1$:

$$\|\mathbf{u}\|^2 = \mathbf{u} \cdot \mathbf{u} = \left(\frac{1}{3}\right)^2 + \left(\frac{-2}{3}\right)^2 + \left(\frac{2}{3}\right)^2 + 0^2 = \frac{1}{9} + \frac{4}{9} + \frac{4}{9} + 0 = 1.$$



Example

Let W be the subspace of $\mathbb{R}^2$ spanned by $\mathbf{x} = (\frac{2}{3}, 1)$. Find a unit vector $\mathbf{z}$ that is a basis for W .



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Let W be the subspace of $\mathbb{R}^2$ spanned by $\mathbf{x} = (\frac{2}{3}, 1)$. Find a unit vector $\mathbf{z}$ that is a basis for W .

W consists of all multiples of $\mathbf{x}$. Any nonzero vector in W is a basis for W . To simplify the calculation, “scale” $\mathbf{x}$ to eliminate fractions. That is, multiply $\mathbf{x}$ by 3 to get

$$\mathbf{y} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}.$$

Compute

$$\|\mathbf{y}\|^2 = 2^2 + 3^2 = 13 \Rightarrow \|\mathbf{y}\| = \sqrt{13}.$$

Normalize $\mathbf{y}$ to get

$$\mathbf{z} = \frac{1}{\sqrt{13}} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} \frac{2}{\sqrt{13}} \\ \frac{3}{\sqrt{13}} \end{pmatrix}.$$

Another unit vector is $\begin{pmatrix} -\frac{2}{\sqrt{13}} \\ \frac{3}{\sqrt{13}} \end{pmatrix}$.



Example (cont.)

Let W be the subspace of $\mathbb{R}^2$ spanned by $\mathbf{x} = (\frac{2}{3}, 1)$. Find a unit vector $\mathbf{z}$ that is a basis for W .

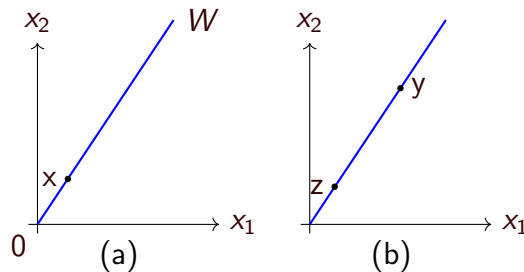


Figure: Normalizing a vector to produce a unit vector.



Definition

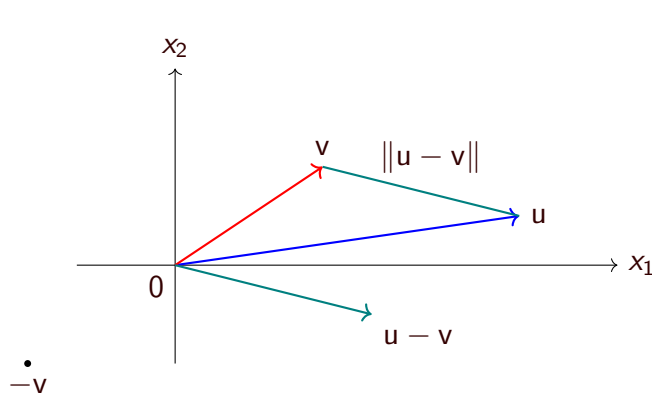
For $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$, the **distance between $\mathbf{u}$ and $\mathbf{v}$** , written as $\text{dist}(\mathbf{u}, \mathbf{v})$ is the length of the vector $\mathbf{u} - \mathbf{v}$. That is,

$$\text{dist}(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \sqrt{(\mathbf{u} - \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v})} = \sqrt{(u_1 - v_1)^2 + \cdots + (u_n - v_n)^2}$$



Example

Compute the distance between the vectors $\mathbf{u} = (7, 1)$ and $\mathbf{v} = (3, 2)$.



- $\mathbf{u} - \mathbf{v} = \begin{pmatrix} 7 \\ 1 \end{pmatrix} - \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \end{pmatrix}.$
- $\sqrt{4^2 + (-1)^2} = \sqrt{17}.$

Figure: The distance between $\mathbf{u}$ and $\mathbf{v}$ is the length of $\mathbf{u} - \mathbf{v}$.



General Inner Products

Definition

An inner product on a vector space V is a real-valued function of vectors $\mathbf{u}$ and $\mathbf{v}$ in V such that for all $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and all scalars c

- a** Symmetry: $(\mathbf{u}, \mathbf{v}) = (\mathbf{v}, \mathbf{u})$
- b** Linearity: $(\mathbf{u} + \mathbf{v}, \mathbf{w}) = (\mathbf{u}, \mathbf{w}) + (\mathbf{v}, \mathbf{w})$
- c** Homogeneity: $(c\mathbf{u}, \mathbf{v}) = c(\mathbf{u}, \mathbf{v}) = (\mathbf{u}, c\mathbf{v})$
- d** Positive definiteness: $(\mathbf{u}, \mathbf{u}) \geq 0$, and $(\mathbf{u}, \mathbf{u}) = 0$ if and only if $\mathbf{u} = \mathbf{0}$



Definition

A **Euclidean space** is a finite-dimensional real vector space V equipped with an inner product $(\cdot, \cdot) : V \times V \rightarrow \mathbb{R}$. The pair $(V, (\cdot, \cdot))$ is called a Euclidean space.



Exercises

Are they inner products?

a $(\mathbf{x}, \mathbf{y}) = 3x_1y_1 + 4x_2y_2$ in $\mathbb{R}^2$

b $(\mathbf{x}, \mathbf{y}) = x_1y_1 - 2x_1y_2 + 3x_2y_1 + 4x_2y_2$ in $\mathbb{R}^2$

c $(\mathbf{x}, \mathbf{y}) = x_1y_1 + x_1y_2 + x_2y_1 + 4x_2y_2 + x_3y_3$ in $\mathbb{R}^3$

d $(A, B) = \det(AB)$ in $V = M_2(\mathbb{R})$

e $(A, B) = \text{trace}(AB)$ in $V = M_2(\mathbb{R})$

f $(A, B) = \text{trace}(A^T B)$ in $V = M_2(\mathbb{R})$



Exercises

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e $(A, B) = \text{trace}(AB)$ in $V = M_2(\mathbb{R})$

f $(A, B) = \text{trace}(A^T B)$ in $V = M_2(\mathbb{R})$

Hint Verify the three defining properties: linearity in the first argument, symmetry, and positive definiteness.



Exercises

Are they inner products?

- a** $(f, g) = f'(t)g'(t)$ in V , all functions differentiable on $[0, 1]$.
- b** $(f, g) = \int_0^1 f(t)g(t)dt$ in V : the vector space of all functions defined on $[0, 1]$.
- c** $(f, g) = \int_0^1 f(t)g(t)dt$ in $V = C[0, 1]$.
- d** $(p, q) = 2p(-1)q(-1) + 3p(0)q(0) + 4p(1)q(1)$ in $V = \mathbb{R}_2[x]$.



Exercises

Are they inner products?

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Hint Verify the three defining properties: linearity in the first argument, symmetry, and positive definiteness.



Inner Products

Definition

Let $\mathbf{u}$, and $\mathbf{v}$ be vectors in V . Then

- Length (or norm) of $\mathbf{u}$: $\|\mathbf{u}\| = \sqrt{(\mathbf{u}, \mathbf{u})}$.
- Unit vector $\|\mathbf{u}\| = 1$.
- Distance between $\mathbf{u}$ and $\mathbf{v}$: $dist(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$.
- Angle $\theta \in [0, \pi]$ is the angle between $\mathbf{u}, \mathbf{v}$: $cos\theta = \frac{(\mathbf{u}, \mathbf{v})}{\|\mathbf{u}\|\|\mathbf{v}\|}$.
- Orthogonality: $\mathbf{u}$, and $\mathbf{v}$ are orthogonal $\iff (\mathbf{u}, \mathbf{v}) = 0$.



Inner Products

If $(\mathbf{x}, \mathbf{y}) = \sum_{ij} a_{ij} x_i y_j$ is an inner product in $\mathbb{R}^n$ then

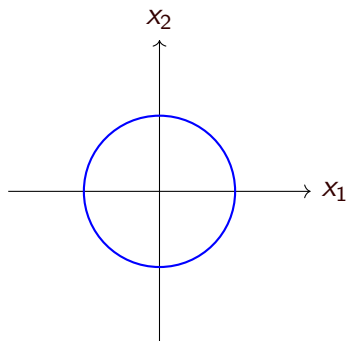
$$(\mathbf{x}, \mathbf{y}) = \mathbf{x}^T R \mathbf{y}$$

where $\mathbf{x}^T = (x_1 \ \dots \ x_n)$, $R = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \vdots & \vdots \\ a_{n1} & \dots & a_{nn} \end{pmatrix}$, $\mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$.

Matrix $R = (a_{ij})$ is called the representation matrix of the inner product.

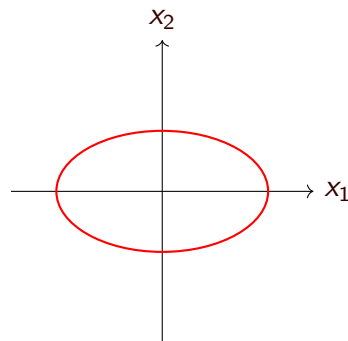
If $U = (\mathbf{x} \ \mathbf{y})$ then $U^T R U = \begin{pmatrix} (\mathbf{x}, \mathbf{x}) & (\mathbf{x}, \mathbf{y}) \\ (\mathbf{y}, \mathbf{x}) & (\mathbf{y}, \mathbf{y}) \end{pmatrix}$. It packages all the geometry (lengths, angles, orthogonality, independence, etc.)





Standard inner product

$$\|x\| = 1 \Rightarrow x_1^2 + x_2^2 = 1$$



General inner product

$$\|x\|_A = 1 \Rightarrow x^T R x = \frac{1}{4}x_1^2 + x_2^2 = 1$$

Figure: Unit sets under the standard inner product (circle) and a general inner product (ellipse).



Standard Inner Products

- On $\mathbb{R}^n$

$$(\mathbf{u}, \mathbf{v}) = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n.$$

- On $C[a, b]$ (continuous real-valued functions)

$$(f, g) = \int_a^b f(t)g(t)dt.$$

- On $M_n(\mathbb{R})$ (real matrices)

$$(A, B) = \text{trace}(A^T B) = \text{trace}(AB^T).$$

Exercises

Determine the norms and distance between $\mathbf{u}$ and $\mathbf{v}$. Are they orthogonal? If not, find the angle between them.

- a** Standard inner product in $\mathbb{R}^3$, $\mathbf{u} = (1, 1, 2)^T$, $\mathbf{v} = (2, 1, -1)^T$.
- b** $(\mathbf{u}, \mathbf{v}) = u_1 v_1 - 2u_1 v_2 - 2u_2 v_1 + 5u_2 v_2$ in $\mathbb{R}^2$, with $\mathbf{u} = (1, 1)^T$, $\mathbf{v} = (0, -1)^T$.
- c** $(\mathbf{u}, \mathbf{v}) = u_1 v_1 + 2u_2 v_2 + 3u_3 v_3 - u_1 v_3 - u_3 v_1$ in $\mathbb{R}^3$, with $\mathbf{u} = (1, 1, 2)^T$, $\mathbf{v} = (2, 1, -1)^T$.



Lemma

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2(\mathbf{u}, \mathbf{v}),$$

and

$$\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2(\mathbf{u}, \mathbf{v}),$$

Theorem (The Pythagorean Theorem)

Two vectors $\mathbf{u}$ and $\mathbf{v}$ in a Euclidean vector spaces V are orthogonal if and only if

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2.$$

Orthogonal Complements

1 Inner Products

2 Orthogonal Complements

3 Orthogonal Projection



Definition

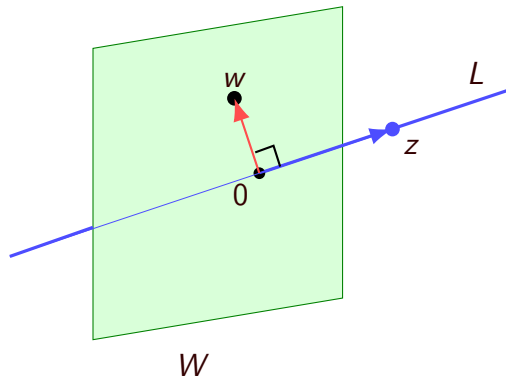
- $\mathbf{u} \perp W$ iff $\mathbf{u} \perp \mathbf{v}$ for all $\mathbf{v} \in W$, where W is a subspace of $\mathbb{R}^n$.
- Orthogonal complement of W : $W^\perp = \{\mathbf{z} \in V : \mathbf{z} \perp W\}$.

$$\begin{aligned} W^\perp &= \{\mathbf{z} \in V : \mathbf{z} \perp W\} \\ &= \{\mathbf{z} \in V : \mathbf{z} \perp \mathbf{v}, \forall \mathbf{v} \in W\} \\ &= \{\mathbf{z} \in V : \mathbf{z} \perp \mathbf{e}, \forall \mathbf{e} \in E\}, E \text{ is a basis of } W. \end{aligned}$$

Theorem

- $\mathbf{u} \in W^\perp$ iff $\mathbf{u} \perp \mathbf{v}$ for all $\mathbf{v} \in W$.
- $W^\perp$ is a subspace of $\mathbb{R}^n$.





A plane and line through 0 as orthogonal complements. $L = W^\perp$ and $W = L^\perp$

Orthogonal Sets

Definition

A set of vectors $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p\}$ in $\mathbb{R}^n$ is said to be an **orthogonal set** if each pair of distinct vectors from the set is orthogonal, that is, if $\mathbf{u}_i \cdot \mathbf{u}_j = 0$ whenever $i \neq j$.

Theorem

If $S = \{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ is an orthogonal set of nonzero vectors in $\mathbb{R}^n$, then S is linearly independent and hence is a basis for the subspace spanned by S .



Example

Are $U = \langle (8, -6, 1), (-7, 5, 0) \rangle$ and $V = \langle (1, 0, -8), (0, 1, 6) \rangle$ orthogonal to each other with the following inner product?

$$(u, v) = u_1 v_1 + 2u_2 v_2 + 3u_3 v_3 - u_1 v_2 - u_2 v_1 + u_2 v_3 + u_3 v_2.$$



Example

Find a and b such that $ax + b$ is orthogonal to $W = \langle x^2, 1 \rangle$ on $[0, 1]$ with the standard inner product.



Orthogonal Complement of Col/Nul

Theorem

Let A be an $m \times n$ matrix and R is the representation matrix of an inner product of a Euclidean vector space $\mathbb{R}^n$.

$$\text{Col}(A)^\perp = \text{Nul}(A^T R) = R^{-1} \text{Nul}(A^T),$$

thus

$$\text{Nul}(A)^\perp = \text{Col}(R^{-1} A^T) = R^{-1} \text{Col}(A^T).$$



Example

Find a basis and the dimension of $F^\perp$ with $F = \langle (1, 1, 1), (2, 1, 0), (1, 0, -1) \rangle$.

a $(\mathbf{x}, \mathbf{y}) = x_1y_1 + 2x_2y_2 + 3x_3y_3 - x_1y_3 - x_3y_1.$

b $(\mathbf{x}, \mathbf{y}) = x_1y_1 + 2x_2y_2 + 3x_3y_3 - x_1y_2 - x_2y_1 + x_2y_3 + x_3y_2.$



Example

Find a basis and the dimension of $F^\perp$ with $F = \langle (1, 1, 1), (2, 1, 0), (1, 0, -1) \rangle$.

a $(\mathbf{x}, \mathbf{y}) = x_1y_1 + 2x_2y_2 + 3x_3y_3 - x_1y_3 - x_3y_1.$

b $(\mathbf{x}, \mathbf{y}) = x_1y_1 + 2x_2y_2 + 3x_3y_3 - x_1y_2 - x_2y_1 + x_2y_3 + x_3y_2.$

Answer

a $F^\perp = \text{span}\{(2, -1, 1)\}, \dim(F^\perp) = 1.$

b $F^\perp = \text{span}\{(-1, -2, 1)\}, \dim(F^\perp) = 1.$



Example

Find a basis and the dimension of $F^\perp$ with

$$F = \{(x_1, x_2, x_3) \in \mathbb{R}_3 : x_1 + x_2 + x_3 = 0, 2x_1 + x_2 - x_3 = 0\}.$$

a $(\mathbf{x}, \mathbf{y}) = x_1y_1 + 2x_2y_2 + 3x_3y_3 - x_1y_3 - x_3y_1.$

b $(\mathbf{x}, \mathbf{y}) = x_1y_1 + 2x_2y_2 + 3x_3y_3 - x_1y_2 - x_2y_1 + x_2y_3 + x_3y_2.$



Example

Find a basis and the dimension of $F^\perp$ with

$$F = \{(x_1, x_2, x_3) \in \mathbb{R}_3 : x_1 + x_2 + x_3 = 0, 2x_1 + x_2 - x_3 = 0\}.$$

a $(\mathbf{x}, \mathbf{y}) = x_1y_1 + 2x_2y_2 + 3x_3y_3 - x_1y_3 - x_3y_1.$

b $(\mathbf{x}, \mathbf{y}) = x_1y_1 + 2x_2y_2 + 3x_3y_3 - x_1y_2 - x_2y_1 + x_2y_3 + x_3y_2.$

Answer $F = \text{span}\{(2, -3, 1)\}, \dim(F) = 1.$

a $\dim(F^\perp) = 3 - \dim(F) = 2, F^\perp = \text{span}\{(6, 1, 0), (-1, 0, 1)\}.$

b $\dim(F^\perp) = 3 - \dim(F) = 2, F^\perp = \text{span}\{(7, 5, 0), (0, 0, 1)\}.$



Example

In $\mathbb{R}^3$, consider the inner product

$(\mathbf{x}, \mathbf{y}) = 4x_1y_1 - 2x_1y_2 + 2x_1y_3 - 2x_2y_1 + 6x_2y_2 + 4x_2y_3 + 2x_3y_1 + 4x_3y_2 + 8x_3y_3$ where $\mathbf{x} = (x_1, x_2, x_3)$, and $\mathbf{y} = (y_1, y_2, y_3)$. Consider the subspace $F = \text{span}\{(1, 0, 1), (-1, 1, 1)\}$.

- a** Find the dimension of $F^\perp$.
- b** Find m such that $\mathbf{u} = (1, -1, m)^T \in F^\perp$.



Example

In $\mathbb{R}^3$, consider the inner product

$(\mathbf{x}, \mathbf{y}) = 4x_1y_1 - 2x_1y_2 + 2x_1y_3 - 2x_2y_1 + 6x_2y_2 + 4x_2y_3 + 2x_3y_1 + 4x_3y_2 + 8x_3y_3$ where $\mathbf{x} = (x_1, x_2, x_3)$, and $\mathbf{y} = (y_1, y_2, y_3)$. Consider the subspace $F = \text{span}\{(1, 0, 1), (-1, 1, 1)\}$.

- a Find the dimension of $F^\perp$.
- b Find m such that $\mathbf{u} = (1, -1, m)^T \in F^\perp$.

Answer

- a $F^\perp = \text{span}\{(-5/4, -5/4, 1)\}$, $\dim(F^\perp) = 1$.
- b $(\mathbf{u}, \mathbf{v}_1) = 10m + 4 = 0$, and $(\mathbf{u}, \mathbf{v}_2) = 10m - 16 = 0 \Rightarrow$ There is no real m .



Example

In $\mathbb{R}^3$, consider the inner product

$(\mathbf{x}, \mathbf{y}) = 4x_1y_1 - 2x_1y_2 + 2x_1y_3 - 2x_2y_1 + 6x_2y_2 + 4x_2y_3 + 2x_3y_1 + 4x_3y_2 + 8x_3y_3$ where $\mathbf{x} = (x_1, x_2, x_3)$, and $\mathbf{y} = (y_1, y_2, y_3)$. Consider the subspace $F = \{(x_1, x_2, x_3) | 2x_1 + 3x_2 - x_3 = 0\}$.

- a** Find the dimension of $F^\perp$.
- b** Find m such that $\mathbf{u} = (1, -1, m)^T \in F^\perp$.



Example

In $\mathbb{R}^3$, consider the inner product

$(\mathbf{x}, \mathbf{y}) = 4x_1y_1 - 2x_1y_2 + 2x_1y_3 - 2x_2y_1 + 6x_2y_2 + 4x_2y_3 + 2x_3y_1 + 4x_3y_2 + 8x_3y_3$ where $\mathbf{x} = (x_1, x_2, x_3)$, and $\mathbf{y} = (y_1, y_2, y_3)$. Consider the subspace $F = \{(x_1, x_2, x_3) | 2x_1 + 3x_2 - x_3 = 0\}$.

- a Find the dimension of $F^\perp$.
- b Find m such that $\mathbf{u} = (1, -1, m)^T \in F^\perp$.

Answer $F = \text{span}\{(0, 1, 3), (1, 0, 2)\}$, $\dim(F) = 2$.

- a $\dim(F^\perp) = 3 - \dim(F) = 1$.
- b $(\mathbf{u}, \mathbf{v}_1) = 28m - 14 = 0$, and $(\mathbf{u}, \mathbf{v}_2) = 18m + 8 = 0 \Rightarrow$ There is no real m .



Orthogonal Projection

- 1 Inner Products
- 2 Orthogonal Complements
- 3 Orthogonal Projection**



Orthogonal Projection

Definition

Let W be a subspace of V . Then each $\mathbf{x} \in V$ can be written uniquely in the form: $\mathbf{x} = \mathbf{y} + \mathbf{z}$ with $\mathbf{y} \in W$ and $\mathbf{z} \in W^\perp$. Besides, $\mathbf{y}$ is called the orthogonal projection of $\mathbf{x}$ on W .

Let $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ be a basis of W and $\{\mathbf{v}_{m+1}, \dots, \mathbf{v}_n\}$ be a basis of $W^\perp$. If V is finite-dimensional, then $\{\mathbf{v}_1, \dots, \mathbf{v}_m, \dots, \mathbf{v}_n\}$ is a basis of V .



General Bases

$\{\mathbf{v}_1, \dots, \mathbf{v}_m, \dots, \mathbf{v}_n\}$ is a basis of V such that $W = \langle \mathbf{v}_1, \dots, \mathbf{v}_m \rangle$. Then $\mathbf{z} = \mathbf{x} - \mathbf{y} \perp W$ with

$$\mathbf{y} = c_1 \mathbf{v}_1 + \dots + c_m \mathbf{v}_m.$$

Thus

$$\sum_{i=1}^m (\mathbf{v}_i, \mathbf{v}_k) c_i = (\mathbf{x}, \mathbf{v}_k).$$

for $k = 1, \dots, m$. That is, if $U = \begin{bmatrix} [\mathbf{v}_1]_E & \dots & [\mathbf{v}_m]_E \end{bmatrix}$, $C = [c_1 \quad \dots \quad c_m]^T$ then

$$(U, U)C = (U, \mathbf{x})$$

with $(U, U) = U^T R U$ and $(U, \mathbf{x}) = U^T R \mathbf{x}$.

Example

Find the orthogonal projection of $\mathbf{x}$ on U .

- a $U = \langle (1, 1, 1), (2, 1, -1) \rangle$, $\mathbf{x} = (1, 2, 3)$ with the inner product $(x, y) = x_1y_1 + 2x_2y_2 + 3x_3y_3 - x_1y_2 - x_2y_1 + x_2y_3 + x_3y_2$.
- b $U = \langle (0, 1, 0, 1), (0, 1, 1, 0), (1, 1, 1, 1), (1, 2, 1, 2) \rangle$, $\mathbf{x} = (1, 1, 0, 0)$ with the inner product $(x, y) = x_1y_1 + 4x_2y_2 + 2x_3y_3 + 3x_4y_4$.



Example

Find the orthogonal projection of $\mathbf{x}$ on U .

- a** $U = \langle (1, 1, 1), (2, 1, -1) \rangle$, $\mathbf{x} = (1, 2, 3)$ with the inner product $(x, y) = x_1y_1 + 2x_2y_2 + 3x_3y_3 - x_1y_2 - x_2y_1 + x_2y_3 + x_3y_2$.
- b** $U = \langle (0, 1, 0, 1), (0, 1, 1, 0), (1, 1, 1, 1), (1, 2, 1, 2) \rangle$, $\mathbf{x} = (1, 1, 0, 0)$ with the inner product $(x, y) = x_1y_1 + 4x_2y_2 + 2x_3y_3 + 3x_4y_4$.

Answer

- a** $y = \text{proj}_U(1, 2, 3) = \frac{17}{7}(1, 1, 1) - \frac{5}{7}(2, 1, -1) = (1, \frac{12}{7}, \frac{22}{7})$.
- b** $y = \text{proj}_U(1, 1, 0, 0) = \frac{7}{25}(0, 1, 0, 1) + \frac{11}{25}(0, 1, 1, 0) + \frac{1}{25}(1, 1, 1, 1) = (\frac{1}{25}, \frac{19}{25}, \frac{12}{25}, \frac{8}{25})$.



Example

Consider the vector space $\mathbb{R}_3$ with the following inner product $(x, y) = x_1y_1 + 2x_2y_2 + 3x_3y_3 - x_1y_2 - x_2y_1 + x_2y_3 + x_3y_2$, and $U = \langle (1, 1, 1), (-1, 1, m) \rangle$.

- a Find m such that U is orthogonal.
- b Let m equal the value obtained above. Determine the orthogonal projection of $\mathbf{x} = (1, 2, 3)$ on U .



Example

Consider the vector space $\mathbb{R}_3$ with the following inner product
 $(x, y) = x_1y_1 + 2x_2y_2 + 3x_3y_3 - x_1y_2 - x_2y_1 + x_2y_3 + x_3y_2$, and
 $U = \langle (1, 1, 1), (-1, 1, m) \rangle$.

- a** Find m such that U is orthogonal.
- b** Let m equal the value obtained above. Determine the orthogonal projection of $x = (1, 2, 3)$ on U .

Answer

- a** $m = \frac{-1}{2}$.
- b** $y = \text{proj}_U x = \frac{8}{3}(1, 1, 1) + \frac{6}{19}(-1, 1, \frac{-1}{2}) = (\frac{134}{57}, \frac{170}{57}, \frac{143}{57})$.



Orthogonal Bases (Bonus)

Definition

An orthogonal basis for a subspace W of $\mathbb{R}^n$ is a basis for W that is also an orthogonal set.

Theorem

In particular, if $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ is orthogonal then

$$\mathbf{y} = c_1 \mathbf{v}_1 + \dots + c_m \mathbf{v}_m,$$

with

$$c_k = \frac{(\mathbf{y}, \mathbf{v}_k)}{(\mathbf{v}_k, \mathbf{v}_k)}, \quad k = 1, \dots, m.$$

Example

Let $\mathbf{u}_1 = (3, 1, 1)^T$, $\mathbf{u}_2 = (-1, 2, 1)^T$, $\mathbf{u}_3 = (-1/2, -2, 7/2)^T$. The set $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal basis for $\mathbb{R}^3$. Express the vector $\mathbf{y} = (6, 1, -8)^T$ as a linear combination of the vectors in S .

